Deleted File Recovery Using Autopsy and FTK Imager

Aim

To recover deleted files from a forensic disk image using FTK Imager and Autopsy and verify the integrity of recovered files using MD5 and SHA-1 hashes.

Requirements

1. FTK Imager
2. Autopsy Digital Forensics Platform
3. Windows computer
4. USB drive or sample disk image (.E01 or .dd)
5. Command Prompt

Algorithm

6. Create a forensic image of the storage device.
7. Verify the acquired image using hash values.
8. Create a new case in Autopsy.
9. Add the forensic image as the data source.
10. Process the image using relevant ingest modules
11. Locate deleted files and recover available data.
12. Calculate hashes for recovered files.
13. Document the recovery results and verification details.

Step-by-Step Procedure

14. A. Create a Forensic Image Using FTK Imager
15. Launch FTK Imager with the required permissions.
16. Select File → Create Disk Image.
17. Choose Physical Drive and select the authorized test USB drive.
18. Select Add and choose the E01 image format.
19. Specify the destination folder, such as C:\Case\01_Images\.
20. Enable image verification if available.
21. Start acquisition and save the generated hash values and acquisition log.

B. Load the Image into Autopsy

22. Launch Autopsy and select New Case.
23. Enter the case name: USB_Deleted_File_Recovery.
24. Set the base directory to C:\Case\02_Working\.
25. Select Add Data Source → Disk Image or VM File.
26. Browse to the acquired .E01 or .dd image.
27. Select the appropriate time zone.

Enable relevant modules, such as File Type Identification, hash lookup, and other available file-recovery or carving modules.

Start processing and wait for ingest to complete.

C. Locate and Recover Deleted Files

28. Expand Views → Deleted Files, or locate deleted-file artifacts in the results.
29. Examine the available deleted-file entries.
30. Select a file whose contents are recoverable.
31. Use the available extraction or export option.
32. Save the recovered file to C:\Case\03_Recovered_Files\.
33. Record its name, size, path, and recovery status.

D. Verify File Integrity

34. Open Command Prompt.
35. Navigate to the recovered-files directory.
36. Calculate the MD5 hash:
37. certutil -hashfile confidential.docx MD5

Calculate the SHA-1 hash:

38. certutil -hashfile confidential.docx SHA1
39. Save the hash outputs in C:\Case\04_Hashes\.
40. Compare the recovered-file hashes with a trusted hash of the original file, if available.

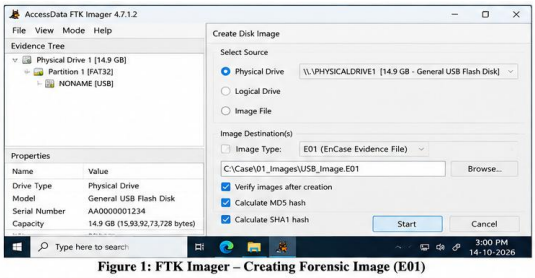


Figure 1: FTK Imager – Creating Forensic Image (E01)

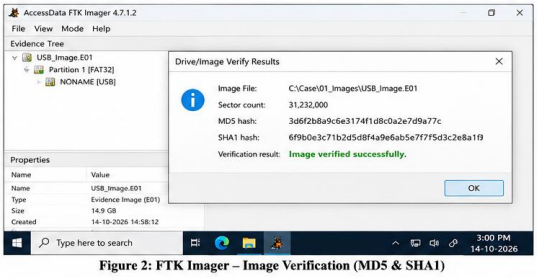


Figure 2: FTK Imager – Image Verification (MD5 & SHA1)

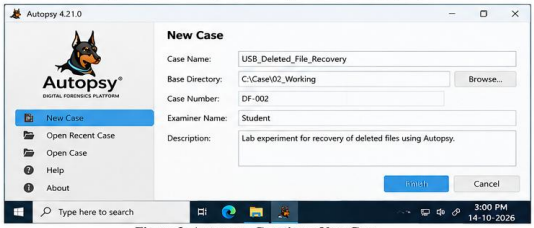


Figure 3: Autopsy – Creating a New Case

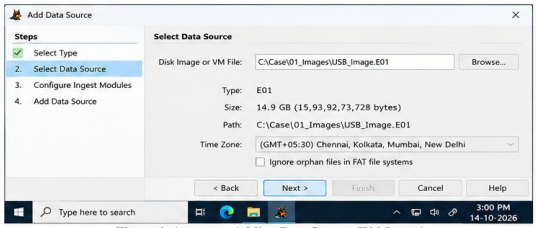


Figure 4: Autopsy – Adding Data Source (E01 Image)

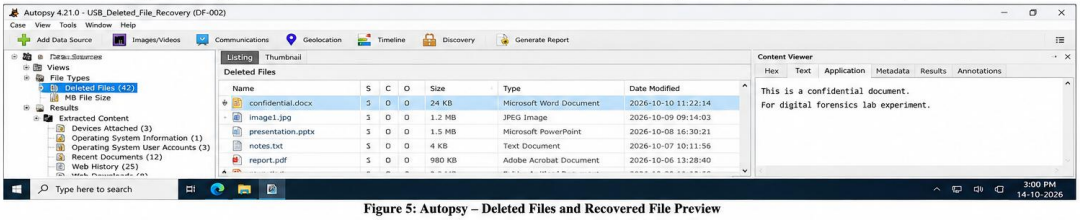


Figure 5: Autopsy – Deleted Files and Recovered File Preview

Result

The procedure for acquiring a forensic image, identifying and recovering available deleted files using Autopsy, and calculating file hashes using Command Prompt was studied. The integrity of recovered files can be assessed by comparing their hashes with trusted original hash values.